SPLUNK LAB REPORT

Home SOC Analyst Lab — Full Setup & Configuration

Date: May 2026 | Platform: Oracle VirtualBox | Status: Active

1. Lab Overview

This report documents the complete setup of a personal cybersecurity home lab using Splunk Enterprise as the SIEM platform. The lab is designed to practice SOC analyst skills including log collection, attack detection, and threat hunting.

1.1 Lab Architecture

Component	Role	IP Address	Status
Host Machine (Windows)	Splunk Enterprise SIEM	Virtual net - 192.168.56.X	Active
Windows 10 VM (VirtualBox)	Target / Victim Machine	Nat Network-192.168.9.X Host - only - 192.168.56.X	Active
Kali Linux VM (VirtualBox)	Attacker Machine	Nat Network- 192.168.9.X	Active

1.2 Network Topology

The lab uses a NAT Network (192.168.9.0/24) inside VirtualBox to isolate the attacker and target VMs from the home network. The host machine sits on the home network (192.168.19.100) and receives logs from the Windows 10 VM via Splunk Universal Forwarder on port 9997.

- Home network: 192.168.19.x (Host Machine)
- Lab NAT Network: 192.168.9.x (Kali + Windows 10 VMs only)
- Kali Linux CANNOT reach the host machine or home router — fully isolated

2. Components Installed

2.1 Splunk Enterprise (Host Machine)

Setting	Value
Splunk Version	Enterprise (Latest)
Web Interface	http://localhost:8000

Receiving Port	9997 (TCP)
Index	main
Firewall Rule	Splunk9997 — Inbound TCP 9997 Allowed

2.2 Splunk Universal Forwarder (Windows 10 VM)

Setting	Value
Installed On	Windows 10 VM (192.168.9.X)
Forward Server	192.168.56.X:9997
Service Account	Local System Account
Service Status	Running
Config File	C:\Program Files\SplunkUniversalForwarder\etc\system\local\inputs.conf

2.3 Sysmon (Windows 10 VM)

Setting	Value
Version	Sysmon v15.20
Config	Olaf Hartong Sysmon Modular Config
Install Path	C:\Sysmon\
Service	Sysmon64 — Running
Events Generated	8,948+ events

3. Log Sources Configured

The following Windows Event Log channels are monitored and forwarded to Splunk:

Source	Splunk Sourcetype	Events Collected	Purpose
Windows Security Log	WinEventLog:Security	20,000+	Login events, auth failures
Windows System Log	WinEventLog:System	Ongoing	System events, services
Windows Application Log	WinEventLog:Application	1,000+	Application events

PowerShell Operational	WinEventLog:Microsoft-Windows-PowerShell/Operational	26+	PowerShell activity
Sysmon Operational	WinEventLog:Microsoft-Windows-Sysmon/Operational	8,948+	Deep process/network monitoring

Total events indexed: 35,905+ events as of lab setup completion.

4. Configuration Files

4.1 inputs.conf (Windows 10 VM Universal Forwarder)

Location: C:\Program Files\SplunkUniversalForwarder\etc\system\local\inputs.conf

Entry	Index	Status
[WinEventLog://Security]	main	Enabled
[WinEventLog://System]	main	Enabled
[WinEventLog://Application]	main	Enabled
[WinEventLog://Microsoft-Windows-Sysmon/Operational]	main	Enabled
[WinEventLog://Microsoft-Windows-PowerShell/Operational]	main	Enabled

5. Troubleshooting & Issues Resolved

Issue	Root Cause	Resolution
0 events in Splunk	Wrong forward server IP	Updated to 192.168.56.X:9997
Ping general failure	VM on NAT, not NAT Network	Switched to NAT Network in VirtualBox
Telnet connect failed	Windows Firewall blocking 9997	Added inbound firewall rule
Forward server inactive	Old IP still configured	Removed old, added correct IP
Sysmon not in Splunk	Wrong channel name (lowercase s)	Fixed to correct case: Sysmon
Sysmon still missing	SplunkForwarder permissions	Set to Local System Account
inputs.conf not working	renderXml causing issues	Removed renderXml, kept simple config

Sysmon install failed	Running from wrong directory	cd C:\Sysmon before install

6. Key Splunk Searches

Search Query	Purpose
index=main | stats count by sourcetype	See all log sources
index=main | stats count by EventCode	See all event types
index=main EventCode=4624	Successful logins
index=main EventCode=4625	Failed login attempts
index=main EventCode=3	Network connections (Sysmon)
index=main EventCode=1	Process creation (Sysmon)
index=main sourcetype="WinEventLog:Microsoft-Windows-Sysmon/Operational"	All Sysmon events
index=_internal source=*metrics.log* group=tcpin_connections	Verify forwarder connections

7. Sysmon Event Code Reference

EventCode	Event Name	Use in Detection
1	Process Created	Detect malicious process execution
3	Network Connection	Detect port scans, C2 connections
7	Image Loaded	Detect DLL injection
11	File Created	Detect malware dropping files
12/13	Registry Modified	Detect persistence mechanisms
22	DNS Query	Detect C2 beaconing

8. Attack Scenarios Planned

#	Attack	Kali Tool	Splunk Detection Query
1	Port Scanning	nmap -sS	index=main EventCode=3 | stats count by SourceIp

2	Failed Login Brute Force	hydra	index=main EventCode=4625 | stats count by TargetUserName
3	Successful Login	Metasploit	index=main EventCode=4624
4	Process Execution	Metasploit payload	index=main EventCode=1 | table Image CommandLine
5	Network Callback (C2)	Meterpreter	index=main EventCode=3 | stats count by DestinationIp

9. Safety & Network Isolation

The lab is designed with security in mind. The following measures ensure the attacks remain isolated:

- Both VMs (Kali and Windows 10) are on a VirtualBox NAT Network (192.168.9.0/24)
- Kali Linux CANNOT reach the host machine (192.168.56.X)
- Kali Linux CANNOT reach the home router or other network devices
- All attack traffic is contained inside VirtualBox
- VM snapshots taken before each attack session for easy recovery
- Windows Firewall enabled on host machine
- No real malware used — only built-in Kali Linux tools

10. Next Steps & Learning Path

Step	Task	Status
1	Splunk Enterprise installation	Completed
2	Universal Forwarder on Windows 10 VM	Completed
3	Windows Event Log collection	Completed
4	Sysmon installation & configuration	Completed
5	Network connectivity between VMs	Completed
6	Kali Linux attack setup	In Progress
7	Nmap scan detection in Splunk	Upcoming
8	Brute force attack & detection	Upcoming
9	Metasploit exploitation & detection	Upcoming
10	Splunk Dashboards & Alerts	Upcoming

This lab is for educational purposes only. All attacks are performed in an isolated virtual environment.